

REMOVING STATIONARITY FROM LOGARITHMIC DLR UNIQUENESS

ABSTRACT. We prove that a canonical one-dimensional logarithmic DLR law at any inverse temperature $\beta > 0$ is translation-invariant whenever the sums of squared unit-interval counts grow at most linearly, almost surely. The hypothesis is pathwise; its random constant need not be integrable. Finite spatial mean energy of one regularized electric field implies this count condition. Consequently the finite-energy uniqueness theorem in the accompanying manuscript extends to laws not assumed stationary, under the same regularization convention. The argument uses compactly supported translations with a logarithmic taper and an exact product identity for two Radon–Nikodym derivatives. A Hellinger estimate avoids expectations of the energy errors. We also record a separate nonstationary $\beta = 2$ consequence of deterministic conditional universality.

Scope. This is a supplement to the supplied manuscript *Uniqueness of the stationary logarithmic DLR state*, file `logarithmic_dlr_uniqueness(2).tex`, denoted by [M] below. The automatic-stationarity theorem and its proof are given in full. Identification with Sine_β then invokes the stationary uniqueness theorem of [M]; this supplement does not reprove that theorem.

1. THE NONSTATIONARY STATEMENTS

Write $\theta_a \gamma = \gamma - a$ and $N_j(\gamma) = \gamma([j, j+1))$, $j \in \mathbb{Z}$. We use the canonical logarithmic conditional densities of Dereudre–Hardy–Leblé–Maïda [?]: condition on both the configuration outside a bounded interval Λ and the number of points inside it. On the ordered configuration simplex the conditional density is positive almost everywhere and proportional to $\exp(-\beta H_\Lambda)$, relative to Lebesgue measure. All exterior energy differences use common symmetric spatial cutoffs. In particular they exist as finite relative energies on the conditional full-measure sets. We retain these Gibbs-kernel equations, without including stationarity in the definition of a solution. This distinction matters because [?, Definition 3.1] packages stationarity and energy assumptions together with the equations.

More explicitly, if T is an increasing C^1 diffeomorphism equal to the identity outside a compact subset of Λ , its energy change is

$$\begin{aligned} \Delta H_T(\gamma) = & - \sum_{\{x,y\} \subset \gamma \cap \Lambda} \log \frac{|T(x) - T(y)|}{|x - y|} \\ & - \lim_{S \rightarrow \infty} \sum_{\substack{x \in \gamma \cap \Lambda \\ y \in \gamma \setminus \Lambda, |y| \leq S}} \log \frac{|T(x) - y|}{|x - y|}. \end{aligned} \tag{1.1}$$

The first sum is over unordered distinct pairs. Normalizing constants and reference configurations disappear in this difference.

Definition 1.1 (A pathwise count condition). A configuration is square-count tempered if

$$M(\gamma) := \sup_{m \in \mathbb{N}, m \geq 1} \frac{1}{m+1} \sum_{|j| \leq m} N_j(\gamma)^2 < \infty. \tag{1.2}$$

Theorem 1.2 (Automatic stationarity). *Fix $\beta > 0$. Let P be a law on locally finite simple configurations on $\mathbb{R}$ satisfying the canonical logarithmic Gibbs-kernel equations described above. If (??) holds P -almost surely, then*

$$P \circ \theta_a^{-1} = P \quad \text{for every } a \in \mathbb{R}.$$

No moment assumption on $M(\gamma)$, no prescribed intensity, and no number-rigidity assumption are required.

For the electric formulation, let ρ_η be uniform probability measure on the circle of radius η in $\mathbb{R}^2$, and regard points of γ as points on the horizontal axis. Consider the pathwise condition that there exist $0 < \eta < 1$ and $F \in L^2_{\text{loc}}(\mathbb{R}^2; \mathbb{R}^2)$ with

$$-\operatorname{div} F = 2\pi(\gamma * \rho_\eta - dx \delta_0), \quad (1.3)$$

$$\limsup_{R \rightarrow \infty} \frac{1}{2R} \int_{[-R, R] \times \mathbb{R}} |F|^2 < \infty. \quad (1.4)$$

Both η and F may depend on the configuration; a measurable selection is unnecessary.

Corollary 1.3 (Finite energy forces stationarity). *A law satisfying the canonical logarithmic Gibbs-kernel equations and supported on configurations satisfying (??)–(??) is stationary.*

Corollary 1.4 (Nonstationary finite-energy uniqueness). *Use the standard inner electric-field truncation and the renormalized energy W specified in [M, Section 7]. Let P , not assumed stationary, be a simple point process satisfying the canonical logarithmic Gibbs-kernel equations, and suppose $\mathbb{E}_P W < \infty$ in the same sense as [M, Theorem 1.2]. Then, using that stationary theorem,*

$$P = \text{Sine}_\beta.$$

The proof of automatic stationarity is a localized-translation argument of the type used for conservation of continuous symmetries; see Georgii [?] and, for the two-dimensional plasma, Leblé [?, Section 8]. Those results concern different settings. We establish all estimates needed for the one-dimensional logarithmic interaction directly below.

2. CUTOFFS WITH VANISHING DEFORMATION COST

For $f \in C^1_c(\mathbb{R})$ define

$$q_f(x, y) = \frac{f(x) - f(y)}{x - y} \quad (x \neq y), \quad q_f(x, x) = f'(x).$$

The diagonal definition makes this a continuous divided difference.

Lemma 2.1 (Deterministic cutoff estimate). *Fix $A \geq 2$. There exist even functions $f_n \in C_c^\infty(\mathbb{R})$, $0 \leq f_n \leq 1$, such that $f_n = 1$ on $[-A, A]$, $\operatorname{supp} f_n \subset [-R_n, R_n]$, $R_n = A2^n$, and*

$$\|f'_n\|_\infty \leq \frac{C}{An}. \quad (2.1)$$

For every configuration satisfying (??),

$$\sum_{\{x, y\} \subset \gamma} q_{f_n}(x, y)^2 + \sum_{x \in \gamma} f'_n(x)^2 \leq \frac{C_{AM}(\gamma)}{n}. \quad (2.2)$$

The sums over pairs are unordered and exclude the diagonal.

Proof. Choose an even smooth function ψ , nonincreasing on $[0, \infty)$, equal to one on $[-1, 1]$ and zero outside $[-2, 2]$. Put

$$f_n(x) = \frac{1}{n} \sum_{k=0}^{n-1} \psi\left(\frac{x}{A2^k}\right). \quad (2.3)$$

The derivative supports lie in the disjoint open annuli $A2^k < |x| < A2^{k+1}$. In particular

$$|f'_n(x)| \leq \frac{C}{n(A + |x|)}, \quad (2.4)$$

$$|f_n(x) - f_n(y)| \leq \frac{C}{n} \left| \log \frac{A + |x|}{A + |y|} \right|. \quad (2.5)$$

The second bound follows by integrating the first along the radial variable and using evenness. The derivative vanishes outside $[-R_n, R_n]$.

Let $I_j = [j, j+1]$ and set

$$d_j = \sup_{I_j} |f'_n|, \quad b_j = \sum_{k \in \mathbb{Z}} \sup_{I_j \times I_k} |q_{f_n}|^2.$$

We claim the row estimates

$$b_j \leq \frac{C_A}{n^2} \begin{cases} (A + |j|)^{-1}, & |j| \leq 4R_n, \\ R_n |j|^{-2}, & |j| > 4R_n. \end{cases} \quad (2.6)$$

Here and below constants may depend on the fixed core size A , but never on n or the configuration.

First, the elementary integral

$$\int_0^\infty \frac{\log^2 t}{(1-t)^2} dt < \infty$$

and the change of variable $t = (A+y)/(A+x)$ show, for $x \geq 0$, that

$$\int_0^\infty \frac{\log^2((A+x)/(A+y))}{(x-y)^2} dy \leq \frac{C}{A+x}.$$

For negative y , replace y by $|y|$ and use $|x-y| \geq |x-|y||$. Reflection covers $x < 0$. Thus

$$\int_{\mathbb{R}} |q_{f_n}(x, y)|^2 dy \leq \frac{C}{n^2(A+|x|)}. \quad (2.7)$$

To pass to cell suprema, neighboring cells are bounded by the mean-value theorem. For $|j-k| \geq 2$, distances between the two cells are comparable to $|j-k|$, and changing either endpoint to the left endpoint of its cell changes f_n by at most d_j or d_k . Comparing the remaining value at k with its average over I_k gives

$$b_j \leq C \int_{\mathbb{R}} |q_{f_n}(j, y)|^2 dy + C \sum_{k \in \mathbb{Z}} \frac{d_j^2 \mathbf{1}_{\{k=j\}} + d_k^2}{(1+|j-k|)^2}. \quad (2.8)$$

For clarity, the d_j^2 errors of all distant cells have already been summed, using $\sum_{h \neq 0} h^{-2} < \infty$. Since $d_k \leq C/[n(A+|k|)]$, splitting the convolution into $|k| \geq |j|/2$ and $|k| < |j|/2$ bounds the second term in (??) by $C_A/[n^2(A+|j|)^2]$. Together with (??), this proves the first row bound (in fact that bound holds for every j).

For the summable far-tail improvement, observe that

$$\sum_{k \in \mathbb{Z}} \sup_{I_k} f_n^2 \leq \frac{C_A R_n}{n^2}. \quad (2.9)$$

Indeed, on the annulus $A2^\ell \leq |x| \leq A2^{\ell+1}$, $f_n(x) \leq (n-\ell)/n$. Summing its squared upper bound times the number of cells gives $C_A n^{-2} \sum_{\ell=0}^{n-1} 2^\ell (n-\ell)^2 \leq C_A R_n/n^2$; the fixed central interval obeys the same bound after increasing C_A . Cells meeting annulus endpoints are covered by enlarging the annuli by one unit. If $|j| > 4R_n$, $f_n = 0$ on I_j and every contributing I_k has distance comparable to $|j|$ from I_j . Equation (??) proves the second row bound.

Put $a_j = N_j(\gamma)^2$. Since $N_j N_k \leq (a_j + a_k)/2$, symmetry of the cell kernel gives

$$\sum_{\{x,y\} \subset \gamma} q_{f_n}(x, y)^2 \leq \frac{1}{2} \sum_j a_j b_j. \quad (2.10)$$

The harmless diagonal overcount only increases the right side. By (??), the sum of the a_j in a dyadic shell of radius $s \geq 1$ is at most $CM(\gamma)s$. Each shell inside $4R_n$ therefore contributes at most $C_A M(\gamma)/n^2$ to (??), and there are $n + O_A(1)$ such shells. Beyond $4R_n$, the shell of radius $2^\ell R_n$ contributes at most $C_A M(\gamma)2^{-\ell}/n^2$. This proves the pair part of (??).

Finally $N_j \leq N_j^2$, because counts are nonnegative integers. The derivative bound on each annulus gives

$$\sum_{x \in \gamma} f'_n(x)^2 \leq \sum_j a_j d_j^2 \leq \frac{C_A M(\gamma)}{n^2},$$

by the same dyadic decomposition, now with summable shell contributions. This completes the proof. $\square$

Remark 2.2. The essential feature is the critical divided-difference kernel $|x - y|^{-2}$. Distributing the change over n dyadic scales costs $O(n^{-1})$. One may also view (??) as the trace of radial cutoffs in $\mathbb{R}^2$ with Dirichlet energy $O(n^{-1})$, that is, vanishing homogeneous $H^{1/2}$ cutoff energy for a fixed interval. The cell estimates above are needed to accommodate arbitrary locally finite configurations.

3. DLR CHANGES OF VARIABLES AND THE HELLINGER ESTIMATE

Proof of Theorem ??. Fix a translation $a \in \mathbb{R}$ and a bounded measurable local observable G , determined by the configuration in a bounded interval B . Choose $A \geq 2$ with $B \subset [-A, A]$, and use Lemma ??. For all sufficiently large n , define

$$T_{n,\pm}(x) = x \pm a f_n(x).$$

By (??) we may assume $|a| \|f'_n\|_\infty \leq 1/2$. These maps are increasing smooth diffeomorphisms, equal to the identity outside $[-R_n, R_n]$. They act on configurations by moving every point. Let

$$Q_{n,\pm} = (T_{n,\pm}^{-1})_* P.$$

We deliberately use the inverse maps in this pushforward.

Choose a bounded interval Λ_n containing $[-R_n, R_n]$ in its interior. Each $T_{n,\pm}$ maps Λ_n onto itself, fixes its exterior, and preserves its number of points. On every conditional fiber of $(\gamma_{\Lambda_n^c}, N_{\Lambda_n})$, the canonical DLR density and ordinary change of variables therefore give

$$w_{n,\pm}(\gamma) := \frac{dQ_{n,\pm}}{dP}(\gamma) = \exp\{-\beta \Delta H_{T_{n,\pm}}(\gamma)\} \prod_{x \in \gamma} (1 \pm a f'_n(x)). \quad (3.1)$$

The product is finite. Positivity of the conditional densities and the diffeomorphism property give equivalence of the measures and $\mathbb{E}_P w_{n,\pm} = 1$. Conditional exceptional Lebesgue-null sets remain null under these diffeomorphisms.

Put $q_n = q_{f_n}$. For every distinct pair x, y ,

$$\frac{|T_{n,+}(x) - T_{n,+}(y)|}{|x - y|} \frac{|T_{n,-}(x) - T_{n,-}(y)|}{|x - y|} = 1 - a^2 q_n(x, y)^2.$$

Applying this identity to the interior pairs and then to the exterior pairs at a common finite cutoff yields

$$w_{n,+} w_{n,-} = e^{-\mathcal{A}_n}, \quad (3.2)$$

where

$$\mathcal{A}_n(\gamma) = -\beta \sum_{\{x,y\} \subset \gamma} \log(1 - a^2 q_n(x, y)^2) - \sum_{x \in \gamma} \log(1 - a^2 f'_n(x)^2). \quad (3.3)$$

There is no uncontrolled linear exterior-force term: it cancels at the common cutoff before the limit is taken. The individual relative energies exist by the specification. The paired infinite sum is absolutely convergent by (??) and the bound below, so passing to the limit preserves (??).

The mean-value theorem gives $|a q_n(x, y)| \leq 1/2$. Using $0 \leq -\log(1 - z) \leq 2z$ for $0 \leq z \leq 1/4$, we obtain, P -almost surely,

$$0 \leq \mathcal{A}_n(\gamma) \leq 2a^2 \left(\beta \sum_{\{x,y\} \subset \gamma} q_n(x, y)^2 + \sum_{x \in \gamma} f'_n(x)^2 \right) \leq \frac{C_{A,a,\beta} M(\gamma)}{n} \rightarrow 0 \quad (3.4)$$

For $Q \ll P$ use the convention $H^2(Q, P) = \int (\sqrt{dQ/dP} - 1)^2 dP$. The arithmetic–geometric mean inequality and (??) give

$$\begin{aligned} H^2(Q_{n,+}, P) + H^2(Q_{n,-}, P) &= 4 - 2\mathbb{E}_P(\sqrt{w_{n,+}} + \sqrt{w_{n,-}}) \\ &\leq 4\mathbb{E}_P[1 - e^{-\mathcal{A}_n/4}] \longrightarrow 0. \end{aligned} \quad (3.5)$$

The last step is bounded convergence. In particular we have not assumed $\mathbb{E}_P M < \infty$, $\mathbb{E}_P \mathcal{A}_n < \infty$, or integrability of individual energy changes. Cauchy–Schwarz gives

$$\|Q - P\|_{\text{TV}} = \frac{1}{2} \int |dQ/dP - 1| dP \leq H(Q, P),$$

so $Q_{n,+} \rightarrow P$ in total variation.

Since $T_{n,+}(x) = x + a$ on B , the restriction of $T_{n,+}^{-1}\gamma$ to B is exactly the restriction of $\theta_a\gamma$ to B . The assertion includes particles entering the interval: $T_{n,+}(B) = B + a$, and the map is one-to-one. Hence

$$\mathbb{E}_{Q_{n,+}} G = \mathbb{E}_P[G \circ \theta_a].$$

The left side tends to $\mathbb{E}_P G$. Equality for every bounded local observable identifies the two probability measures, and a was arbitrary. This proves stationarity. $\square$

4. WHY FINITE ELECTRIC ENERGY SUPPLIES THE COUNT CONDITION

Lemma 4.1. *Every locally finite configuration satisfying (??)–(??) satisfies (??).*

Proof. Choose $\chi \in C_c^\infty((-2, 3) \times (-2, 2))$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-1, 2] \times [-1, 1]$. Write $\chi_j(x, y) = \chi(x - j, y)$ and $S_j = [j - 2, j + 3] \times [-2, 2]$. Since $0 < \eta < 1$, every smearing circle whose center lies in $[j, j + 1) \times \{0\}$ is contained in $\{\chi_j = 1\}$. Positivity and the divergence identity give

$$\begin{aligned} N_j &\leq \int \chi_j d(\gamma * \rho_\eta) \\ &= \int_{\mathbb{R}} \chi_j(x, 0) dx + \frac{1}{2\pi} \int_{\mathbb{R}^2} F \cdot \nabla \chi_j \\ &\leq 5 + C \left(\int_{S_j} |F|^2 \right)^{1/2}. \end{aligned}$$

After squaring and summing, the boxes S_j have bounded overlap, so

$$\sum_{|j| \leq m} N_j^2 \leq C(m + 1) + C \int_{[-m-3, m+3] \times \mathbb{R}} |F|^2. \quad (4.1)$$

The right side is $O_\gamma(m)$ by (??); the finitely many remaining small m have finite counts. This proves the lemma. $\square$

Proof of Corollaries ?? and ??. Lemma ?? and Theorem ?? prove the first corollary. For the second, recall the convention of [M]:

$$f_\eta(z) = \mathbf{1}_{\{|z| \leq \eta\}} \log(|z|/\eta), \quad E_\eta = E + \sum_{p \in \gamma} \nabla f_\eta(\cdot - (p, 0)).$$

If $-\operatorname{div} E = 2\pi(\gamma - dx \delta_0)$, then $F = E_\eta$ satisfies (??).

Finiteness of the renormalized energy, defined as the infimum over compatible fields of their small-truncation renormalized spatial mean energy, provides a compatible field for which this functional is finite. The infimum need not be attained. For at least one sufficiently small fixed η , its spatial mean of $|E_\eta|^2$ is finite: the self-energy subtraction at fixed η is a finite constant. This is exactly the fixed-truncation implication used in [M, Lemma 7.2]. It is pathwise and requires no stationarization. Thus (??)–(??) hold P -almost surely.

The preceding corollary proves that P is stationary. Its assumed $\mathbb{E}_P W < \infty$ and its canonical DLR equations are unchanged. It now satisfies all hypotheses of [M, Theorem 1.2], and hence $P = \text{Sine}_\beta$. $\square$

Remark 4.2 (The precise energy assumption). The inner truncation indicator is essential; we use the convention already specified in [M], rather than the outer indicator printed in the supplied version of [?, Definition 2.20]. For an alternative notion of “finite energy”, the implication (??)–(??) must be checked. It is not enough merely to produce a stationary spatial average of the candidate law.

5. WHAT CHANGES IN THE ACCOMPANYING MANUSCRIPT

The argument supplies a preliminary stationarity theorem. The gap-entropy comparison, random particle-ended blocks, Palm identities, and stationary electric lift in [M] can subsequently be used as written. They are applied only after stationarity has been proved.

In particular, the finite-energy theorem can be restated as follows:

Fix $\beta > 0$, with the inner-truncation energy convention of [M]. Every simple point-process law satisfying the canonical logarithmic Gibbs-kernel equations and $\mathbb{E}_P W < \infty$ equals Sine_β .

Its new proof first invokes Corollary ?? and then the original stationary theorem. The formulation must refer to the Gibbs-kernel equations themselves, rather than invoking a definition which already assumes stationarity.

For the discrepancy theorem, a similarly useful sufficient condition is (??) together with the three count bounds in [M], with their variance function replaced by $d_P(t) = \mathbb{E}_P(N_{(0,t]} - t)^2$. Theorem ?? first supplies stationarity; the linear centered bound then forces intensity one, and the original comparison theorem applies.

The reason this additional argument is needed is structural. Averages of many gap observables cannot distinguish a law from another law with different behavior in a fixed window but the same spatial averages. Stationarity is used when [M] identifies the limiting averages with a Palm gap distribution. The present proof directly controls translation of a fixed window and therefore resolves that missing step. No assertion about a local defect being a DLR solution is needed for this observation.

6. AN INDEPENDENT NONSTATIONARY RESULT AT $\beta = 2$

Proposition 6.1. *Let P satisfy the canonical logarithmic Gibbs-kernel equations at $\beta = 2$. Assume that, almost surely, its points admit an increasing two-sided enumeration $(p_k)_{k \in \mathbb{Z}}$ with*

$$p_k/k \longrightarrow 1 \quad (k \rightarrow \pm\infty), \quad \lim_{S \rightarrow \infty} \sum_{\substack{p \in \gamma \\ 0 < |p| \leq S}} \frac{1}{p} \text{ exists.}$$

Then $P = \text{Sine}_2$, without stationarity or an energy assumption.

Proof. Fix a configuration satisfying these assumptions. For the interval $[-R, R]$ let $N_R = \gamma([-R, R])$. Its canonical conditional law is the N_R -particle orthogonal-polynomial ensemble with weight

$$w_R(x) = \mathbf{1}_{[-R, R]}(x) \lim_{S \rightarrow \infty} \prod_{\substack{p \in \gamma \\ R < |p| \leq S}} \left(1 - \frac{x}{p}\right)^2.$$

Kuijlaars–Miña-Díaz [?, Assumption 1.2 and Theorem 1.3] prove that the associated kernels K_R converge locally uniformly to

$$K_{\sin}(x, y) = \frac{\sin \pi(x - y)}{\pi(x - y)}, \quad K_{\sin}(x, x) = 1.$$

Their particle number is exactly the actual count N_R above.

For $h \in C_c^+(\mathbb{R})$ supported in a compact interval B , the conditional Laplace functional has the Fredholm series

$$L_R(\gamma) = \sum_{k=0}^{N_R} \frac{(-1)^k}{k!} \int_{B^k} \prod_{i=1}^k (1 - e^{-h(x_i)}) \det[K_R(x_i, x_j)]_{i,j=1}^k d\mathbf{x}.$$

For each fixed admissible configuration, eventually $\sup_{x \in B} K_R(x, x) \leq 2$. Positive semidefiniteness and Hadamard's inequality bound the absolute k th summand by $(2|B|)^k/k!$. Local uniform convergence and dominated convergence in the series therefore yield $L_R(\gamma) \rightarrow \mathbb{E}_{\text{Sine}_2} e^{-\langle h, \gamma \rangle}$. Finally DLR gives

$$\mathbb{E}_P e^{-\langle h, \gamma \rangle} = \mathbb{E}_P L_R(\gamma),$$

and $0 \leq L_R \leq 1$ allows bounded convergence over P . Equality of all such Laplace functionals proves the proposition. The random value of N_R causes no difficulty: the deterministic theorem is applied separately to each full configuration. $\square$

This last result provides a separate check on the nonstationary conclusion at $\beta = 2$. The all- β conclusion above instead comes from the explicit automatic-stationarity argument and the stationary theorem in [M].

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